

Lecture 2.2: Tumor Suppressors - Gatekeepers and Guardians

Good morning, everyone. Today we're embarking on a journey through one of the most fascinating and clinically important aspects of cancer biology: tumor suppressor genes. If oncogenes are the accelerator pedal in cancer development, then tumor suppressors are the brakes—the cellular mechanisms that normally prevent malignant transformation.

Over the next three hours, we'll explore three of the most important tumor suppressors: TP53, often called "the guardian of the genome," APC, the gatekeeper of the Wnt signaling pathway, and RB, the master controller of cell cycle progression. Each of these proteins tells a unique story about how cells protect themselves from cancer, and how the loss of these protections leads to malignancy.

What makes tumor suppressors particularly compelling is their dual nature. They're essential for normal cellular function, yet their loss or inactivation is one of the most common features of human cancer. Understanding how they work—and how they fail—has been crucial for both our basic understanding of cancer and for developing new therapeutic approaches.

The Concept of Tumor Suppression: A Historical Perspective

Before we dive into specific tumor suppressors, let's establish the conceptual framework. The idea of tumor suppressor genes emerged from Knudson's two-hit hypothesis, which we discussed in our historical overview. Unlike oncogenes, which cause cancer when activated, tumor suppressors cause cancer when they're lost or inactivated.

This creates an important asymmetry in cancer biology. Oncogenes are typically activated by gain-of-function mutations—changes that make them more active or stable. Tumor suppressors are typically inactivated by loss-of-function mutations—changes that eliminate or reduce their protective functions.

This asymmetry has profound implications for how these genes contribute to cancer and how we might target them therapeutically. Activating a lost tumor suppressor is much more challenging than inhibiting an overactive oncogene.

TP53: The Guardian of the Genome

Let's begin with TP53, arguably the most important tumor suppressor gene in human cancer. TP53 is mutated in over 50% of all human cancers, making it the most frequently altered gene in cancer. But its importance goes far beyond its mutation frequency—TP53 sits at the center of a vast network of cellular protective mechanisms.

The Guardian Metaphor: Why TP53 Earned Its Title

TP53 has been called "the guardian of the genome" because of its central role in protecting cellular DNA integrity. Think of TP53 as a sophisticated security system that monitors cellular health and responds to threats.

When DNA is damaged, when oncogenes become activated, when cells experience stress that might lead to malignant transformation, TP53 springs into action. It can halt cell division to allow time for repair, activate DNA repair mechanisms, or if the damage is too severe, trigger cellular suicide through apoptosis.

This makes TP53 one of our most important defenses against cancer. It's the protein that eliminates potentially dangerous cells before they can become malignant. The fact that TP53 is so frequently mutated in cancer tells us just how crucial this guardian function is—cancer cells must disable this security system to survive and proliferate.

Molecular Functions: A Multi-Talented Guardian

TP53 is a transcription factor, but calling it simply a transcription factor understates its sophistication. It's more accurately described as a stress-responsive transcriptional regulator that coordinates multiple cellular responses to protect against malignant transformation.

DNA Damage Detection and Response

TP53 doesn't directly detect DNA damage—it responds to signals from DNA damage sensors throughout the cell. When proteins like ATM and ATR detect DNA double-strand breaks, or when other sensors detect various types of DNA lesions, they send signals that stabilize and activate TP53.

Once activated, TP53 can initiate several different responses depending on the severity of the damage and the cellular context. For minor damage that can likely be repaired, TP53 induces temporary cell cycle arrest, giving the cell time to fix the problem. For more severe damage, TP53 can trigger permanent cell cycle arrest (senescence) or apoptosis.

This decision-making process is remarkably sophisticated. TP53 somehow integrates information about the extent of DNA damage, the cell's repair capacity, the cellular environment, and other factors to determine the most appropriate response. We're still working to understand exactly how this molecular decision-making process works.

Cell Cycle Checkpoint Control

TP53 controls two major cell cycle checkpoints: the G1/S checkpoint and the G2/M checkpoint. The G1/S checkpoint prevents cells with damaged DNA from replicating that damage, while the G2/M checkpoint prevents cells from dividing until DNA repair is complete.

At the G1/S checkpoint, TP53 activation leads to the transcription of p21, a protein that binds to and inhibits cyclin-dependent kinases. This effectively puts the brakes on cell cycle progression, preventing the cell from entering S phase and replicating its DNA until the damage is repaired.

The G2/M checkpoint involves a more complex network of proteins, but TP53 plays a crucial role here as well. By preventing damaged cells from dividing, TP53 ensures that DNA damage isn't passed on to daughter cells.

Apoptosis Induction

When DNA damage is too extensive to repair, or when other cellular stress signals indicate that a cell has become dangerous, TP53 can trigger apoptosis—programmed cell death. This might seem drastic, but it's actually one of our most important cancer prevention mechanisms.

TP53 induces apoptosis by activating the transcription of pro-apoptotic genes like BAX and PUMA. These proteins disrupt mitochondrial function, leading to the release of cytochrome c and the activation of caspases—the enzymes that execute the apoptotic program.

The decision between cell cycle arrest and apoptosis appears to depend on multiple factors, including the extent of DNA damage, the cell type, the presence of survival signals, and the cellular context. This decision-making process is still an active area of research.

Regulatory Networks: Controlling the Guardian

Given TP53's powerful effects on cell survival, its activity must be tightly regulated. Under normal conditions, TP53 protein levels are kept very low through rapid degradation. Only when stress signals indicate a need for TP53 function does the protein accumulate to significant levels.

MDM2-Mediated Regulation

The primary regulator of TP53 under normal conditions is MDM2, an E3 ubiquitin ligase that targets TP53 for degradation. MDM2 binds directly to TP53 and tags it with ubiquitin, marking it for destruction by the proteasome.

This creates an elegant regulatory circuit. Under normal conditions, MDM2 keeps TP53 levels low, preventing unnecessary cell cycle arrest or apoptosis. But when stress signals activate TP53, the protein can transcriptionally activate MDM2, creating a negative feedback loop that helps return TP53 levels to normal once the stress is resolved.

DNA Damage-Induced Stabilization

When cells experience DNA damage or other stress, multiple kinases phosphorylate TP53 at sites that interfere with MDM2 binding. This stabilizes TP53, allowing it to accumulate and activate its target genes.

The kinases involved include ATM and ATR (activated by DNA damage), Chk1 and Chk2 (DNA damage checkpoint kinases), and others. This creates a robust system where multiple types of cellular stress can activate TP53 through overlapping but distinct pathways.

Target Genes: The TP53 Response Network

TP53 regulates the transcription of hundreds of genes, creating a complex network of cellular responses. Let me highlight some of the most important categories:

Cell Cycle Arrest Genes

p21 is perhaps the most important TP53 target for cell cycle control. p21 protein binds to and inhibits cyclin-dependent kinases, particularly the cyclin E-CDK2 and cyclin D-CDK4/6 complexes that drive G1/S transition. This creates a molecular brake that prevents cells from entering S phase.

But p21 isn't the only cell cycle target. TP53 also regulates genes encoding other CDK inhibitors, proteins involved in DNA replication control, and cell cycle checkpoint proteins.

Apoptosis Genes

BAX and PUMA are key pro-apoptotic targets of TP53. BAX is a member of the BCL2 family that promotes mitochondrial outer membrane permeabilization, leading to cytochrome c release and apoptosome formation. PUMA is a BH3-only protein that activates BAX and inhibits anti-apoptotic BCL2 family members.

TP53 also regulates other apoptosis genes, including death receptors like FAS, pro-apoptotic BCL2 family members like NOXA, and proteins involved in the intrinsic apoptotic pathway.

DNA Repair Genes

Somewhat paradoxically, TP53 also activates genes involved in DNA repair. This makes biological sense—if a cell is going to survive DNA damage, TP53 wants to ensure that the damage is properly repaired.

TP53 targets include genes involved in nucleotide excision repair, base excision repair, homologous recombination, and other DNA repair pathways. This helps explain why TP53 activation can sometimes promote cell survival rather than cell death.

Cancer Significance: When the Guardian Fails

The importance of TP53 in cancer prevention is dramatically illustrated by what happens when it's lost. TP53 mutations are found in over 50% of human cancers, making it the most frequently mutated gene in cancer.

Mutation Patterns and Consequences

Most TP53 mutations in cancer are missense mutations that produce altered proteins rather than eliminating the protein entirely. Many of these mutations occur in the DNA-binding domain and eliminate TP53's ability to regulate its target genes.

Some TP53 mutations have what are called "gain-of-function" effects—they not only eliminate normal TP53 function but also give the mutant protein new, cancer-promoting activities. These mutant proteins can interfere with other tumor suppressor pathways and promote cellular transformation.

The loss of TP53 function has profound effects on cellular behavior. Cells become resistant to apoptosis, can bypass cell cycle checkpoints, and can tolerate much higher levels of DNA damage. This enables the genomic instability that drives cancer progression.

Li-Fraumeni Syndrome

The importance of TP53 is also illustrated by Li-Fraumeni syndrome, a rare hereditary cancer syndrome caused by germline TP53 mutations. Individuals with this syndrome develop multiple cancers at young ages, often including sarcomas, breast cancers, brain tumors, and adrenal cortical carcinomas.

Li-Fraumeni syndrome provides a natural experiment in what happens when TP53 function is compromised from birth. The fact that these individuals develop cancer so early and so frequently demonstrates just how crucial TP53 is for cancer prevention.

Therapeutic Restoration: Bringing Back the Guardian

The frequency of TP53 mutations in cancer has made it an attractive target for therapeutic intervention. However, restoring function to a mutated tumor suppressor is much more challenging than inhibiting an overactive oncogene.

Small Molecule Approaches

Several strategies are being pursued to restore TP53 function. One approach involves small molecules that can bind to mutant TP53 proteins and restore their normal conformation and function. Compounds like PRIMA-1 and APR-246 have shown promise in preclinical studies and early clinical trials.

Another approach targets the MDM2-TP53 interaction in tumors that retain wild-type TP53. Drugs like nutlin-3 and its analogs can block MDM2 binding to TP53, stabilizing the protein and activating TP53-dependent responses even in the absence of DNA damage.

Gene Therapy Approaches

Direct gene therapy to introduce wild-type TP53 into tumor cells has also been explored. While technically challenging, some approaches using viral vectors to deliver functional TP53 have shown clinical activity.

Synthetic Lethality

An alternative approach exploits the concept of synthetic lethality. Tumors with TP53 mutations often become dependent on alternative survival pathways. Identifying and targeting these dependencies can selectively kill TP53-mutant cells while sparing normal cells with functional TP53.

APC: The Wnt Pathway Gatekeeper

Now let's turn our attention to APC, the adenomatous polyposis coli protein. While less universally important than TP53, APC is crucial for specific cancer types, particularly colorectal cancer. APC mutations are found in approximately 80% of colorectal cancers, making it the most important tumor suppressor for this cancer type.

The Gatekeeper Concept

APC has been called a "gatekeeper" tumor suppressor because it controls entry into the neoplastic state in colorectal epithelium. Unlike TP53, which responds to cellular stress, APC functions continuously to maintain normal cellular behavior. Its loss is often the first step in colorectal tumorigenesis.

The gatekeeper function of APC is closely tied to its role in the Wnt signaling pathway, one of the most important developmental and homeostatic signaling pathways in multicellular organisms. When APC function is lost, Wnt signaling becomes constitutively active, driving the initial steps of colorectal cancer development.

The Destruction Complex: APC's Central Role

To understand how APC functions as a tumor suppressor, we need to understand the Wnt signaling pathway and APC's central role in controlling it.

The β -Catenin Destruction Complex

Under normal conditions, APC is the central component of a protein complex whose job is to keep β -catenin levels low. β -catenin is a key effector of Wnt signaling—when β -catenin levels are low, Wnt signaling is off; when β -catenin accumulates, Wnt signaling is active.

The destruction complex includes APC, Axin, glycogen synthase kinase 3 beta (GSK-3 β), and casein kinase 1 (CK1). APC serves as a scaffold that brings together the other components and positions them to phosphorylate β -catenin.

The process works like this: APC and Axin form the core of the complex, GSK-3 β and CK1 phosphorylate β -catenin at specific serine and threonine residues, and these phosphorylations mark β -catenin for ubiquitination and degradation.

Wnt Signal Transduction

When Wnt ligands bind to their receptors, they initiate a signaling cascade that ultimately inactivates the destruction complex. This allows β -catenin to accumulate, translocate to the nucleus, and activate transcription of Wnt target genes.

The Wnt target genes include many that promote cell proliferation, block differentiation, and enhance cell survival. In the intestinal epithelium, appropriate Wnt signaling is essential for maintaining the stem cell compartment and driving the continuous renewal of the intestinal lining.

APC's Multiple Functions

APC has multiple domains that allow it to interact with different components of the destruction complex and to perform additional functions beyond β -catenin regulation. It has binding sites for β -catenin, Axin, microtubules, and other proteins.

This multi-domain structure allows APC to serve as a platform for organizing protein complexes and coordinating different aspects of cellular behavior.

Cytoskeletal Functions: Beyond Wnt Signaling

While APC's role in Wnt signaling is its most important tumor suppressor function, APC also has significant effects on the cytoskeleton that contribute to its overall cellular functions.

Microtubule Dynamics

APC binds directly to microtubules and affects their dynamics. It promotes microtubule stabilization and organization, particularly at the cell cortex. This function is important for maintaining cell polarity and for proper chromosome segregation during cell division.

Loss of APC function can lead to chromosomal instability, partly through its effects on microtubule organization. This contributes to the genomic instability that characterizes many colorectal cancers.

Cell Polarity and Migration

APC is important for establishing and maintaining cell polarity, particularly in epithelial cells. It helps organize the cytoskeleton in a way that establishes distinct apical and basolateral domains.

APC also affects cell migration through its effects on both the Wnt pathway and the cytoskeleton. Loss of APC function can enhance cell motility and contribute to the invasive behavior of cancer cells.

Cancer Role: The Colorectal Connection

APC's role in colorectal cancer is so fundamental that it's hard to overstate. APC mutations are found in the earliest detectable colorectal lesions and in the vast majority of advanced colorectal cancers.

The Adenoma-Carcinoma Sequence

Colorectal cancer typically develops through a well-characterized sequence of changes known as the adenoma-carcinoma sequence. APC mutation is typically the first step in this sequence, leading to the formation of small adenomatous polyps.

These early APC-mutant lesions show activated Wnt signaling, increased cell proliferation, and reduced apoptosis. While they're not yet malignant, they represent the first step toward cancer and provide the cellular context in which additional mutations can accumulate.

As additional mutations accumulate—typically in genes like KRAS, PIK3CA, and TP53—these benign adenomas progress to invasive carcinomas. But APC mutation remains present throughout this progression, highlighting its fundamental importance.

Familial Adenomatous Polyposis

The importance of APC is also illustrated by familial adenomatous polyposis (FAP), a hereditary syndrome caused by germline APC mutations. Individuals with FAP develop hundreds to thousands of colorectal polyps beginning in their teens, and without intervention, virtually all will develop colorectal cancer by age 40.

FAP demonstrates what happens when APC function is compromised from birth in the colorectal epithelium. The massive polyposis that develops shows how crucial APC is for maintaining normal intestinal homeostasis.

Mutation Patterns

Most APC mutations in colorectal cancer are nonsense or frameshift mutations that produce truncated proteins. These truncated proteins typically lose their β -catenin binding and regulation functions while retaining some other functions.

Interestingly, different positions of APC mutations can produce different severities of Wnt pathway activation. This suggests that the precise nature of APC loss can influence the biological behavior of the resulting tumor.

Therapeutic Targeting: Modulating Wnt Signaling

The central role of APC loss in colorectal cancer has made the Wnt pathway an attractive target for therapeutic intervention. However, developing effective Wnt pathway inhibitors has proven challenging.

Wnt Signaling Modulators

Several approaches are being pursued to inhibit Wnt signaling in APC-mutant cancers. These include small molecules that target different components of the pathway, including β -catenin itself, the transcriptional machinery that β -catenin activates, and the enzymes that regulate β -catenin stability.

One challenge is that Wnt signaling is important for normal stem cell function in many tissues. This means that systemic Wnt inhibition can have significant toxicity, particularly affecting tissues with high turnover like the intestine and bone marrow.

β -Catenin/TCF Inhibitors

A more targeted approach involves disrupting the interaction between β -catenin and TCF/LEF transcription factors, which mediate β -catenin's transcriptional effects. Small molecules that block this protein-protein interaction could potentially inhibit Wnt signaling without affecting β -catenin's other functions.

Targeting Wnt Target Genes

Another approach involves targeting the downstream effectors of Wnt signaling rather than the pathway itself. If specific Wnt target genes are particularly important for cancer cell survival or proliferation, inhibiting these targets might be more selective than blocking the entire pathway.

RB: The Cell Cycle Control Master

Our final tumor suppressor, RB (retinoblastoma protein), was actually the first tumor suppressor gene to be discovered and molecularly characterized. RB's story illustrates many of the fundamental principles of tumor suppressor biology and provides crucial insights into how cell cycle control goes wrong in cancer.

Historical Significance: The First Tumor Suppressor

RB holds a special place in cancer biology history as the tumor suppressor that validated Knudson's two-hit hypothesis. The statistical patterns that Knudson observed in retinoblastoma incidence were eventually explained by the molecular characterization of the RB gene and protein.

But RB's importance extends far beyond its historical significance. RB is inactivated in the majority of human cancers, either through direct mutation, deletion, viral protein interaction, or functional inactivation through other mechanisms.

E2F Regulation: The Heart of RB Function

RB's primary function is to control the G1/S transition of the cell cycle by regulating E2F transcription factors. Understanding this regulatory mechanism is crucial for appreciating how RB prevents cancer and how its loss contributes to malignant transformation.

The E2F Family

The E2F family consists of eight transcription factors (E2F1 through E2F8) that have diverse roles in cell cycle control. E2F1, E2F2, and E2F3 are the primary targets of RB regulation and are crucial for S-phase gene expression.

These E2F proteins regulate the transcription of genes encoding DNA replication machinery, nucleotide biosynthesis enzymes, cell cycle regulatory proteins, and many other factors needed for DNA synthesis and cell division.

RB-E2F Interaction

In its active, hypophosphorylated state, RB binds directly to E2F1, E2F2, and E2F3, sequestering them and preventing them from activating transcription. This binding is not simply inhibitory—RB actively recruits chromatin-modifying enzymes that repress E2F target genes.

The RB-E2F interaction is regulated by cell cycle-dependent phosphorylation of RB. During G1, cyclin D-CDK4/6 complexes begin phosphorylating RB at multiple sites. As cells progress toward S phase, cyclin E-CDK2 complexes complete RB phosphorylation.

Hyperphosphorylated RB releases E2F, allowing these transcription factors to activate S-phase gene expression and drive DNA replication.

Chromatin Modification: Active Repression

RB doesn't just passively sequester E2F—it actively represses E2F target genes through chromatin modifications.

HDAC Recruitment

RB recruits histone deacetylases (HDACs) to E2F target gene promoters. HDACs remove acetyl groups from histones, creating a chromatin structure that's generally associated with transcriptional repression.

This active repression helps ensure that S-phase genes remain firmly off during G1, providing a robust block to inappropriate cell cycle progression.

Other Chromatin Modifiers

RB also interacts with other chromatin-modifying enzymes, including histone methyltransferases and chromatin remodeling complexes. This creates a comprehensive repressive chromatin environment at E2F target genes.

The recruitment of multiple chromatin modifiers suggests that RB-mediated repression is meant to be robust and difficult to overcome accidentally.

Differentiation Control: Beyond Cell Cycle

While RB is best known for its cell cycle functions, it also plays crucial roles in cellular differentiation. In many cell types, terminal differentiation requires permanent cell cycle exit, and RB is essential for establishing and maintaining this differentiated state.

Differentiation Programs

RB is required for the terminal differentiation of multiple cell types, including muscle cells, neurons, and adipocytes. In these contexts, RB appears to work both by blocking cell cycle progression and by actively promoting differentiation gene expression.

The mechanism involves RB's interaction with tissue-specific transcription factors that drive differentiation programs. RB can serve as a co-activator for these factors while simultaneously blocking E2F-dependent proliferation genes.

Differentiation vs. Proliferation

The relationship between differentiation and proliferation is complex, but in general, terminally differentiated cells cannot proliferate. RB helps enforce this relationship by creating a molecular switch that couples cell cycle exit with differentiation gene activation.

Loss of RB function can therefore disrupt both cell cycle control and differentiation, contributing to the dedifferentiated appearance of many cancers.

Clinical Relevance: Universal Cancer Connection

RB is inactivated in the vast majority of human cancers, making it one of the most broadly important tumor suppressors.

Mechanisms of Inactivation

RB can be inactivated through several mechanisms:

- Direct mutation or deletion of the RB gene
- Viral protein interaction (as with HPV E7 protein)
- Functional inactivation through overexpression of cyclins or CDKs
- Loss of CDK inhibitors like p16INK4A
- Amplification of CDK4 or CDK6

This diversity of inactivation mechanisms reflects the crucial importance of RB function for cancer prevention. Cancer cells must find some way to overcome RB-mediated cell cycle control.

Therapeutic Implications

The universal importance of RB pathway inactivation in cancer has made this pathway an attractive target for therapeutic intervention. CDK4/6 inhibitors like palbociclib, ribociclib, and abemaciclib have shown significant clinical activity, particularly in breast cancer.

These drugs work by blocking the kinases that phosphorylate and inactivate RB, effectively restoring RB function even in cancers where RB itself is not mutated.

Integration: Tumor Suppressors in the Hallmarks Framework

As we conclude our exploration of these three crucial tumor suppressors, let's consider how they fit into the hallmarks of cancer framework we discussed earlier.

Multiple Hallmark Connections

Each tumor suppressor we've discussed affects multiple cancer hallmarks:

TP53 primarily affects "resisting cell death" and "enabling replicative immortality" through its roles in apoptosis and senescence. But it also affects "genome instability and mutation" through its DNA damage response functions.

APC primarily affects "sustaining proliferative signaling" and "evading growth suppressors" through its role in Wnt pathway regulation. But it also affects "activating invasion and metastasis" through its cytoskeletal functions.

RB primarily affects "evading growth suppressors" through its cell cycle control functions. But it also affects cellular differentiation programs that influence multiple other hallmarks.

Therapeutic Opportunities

Understanding tumor suppressor functions has opened up several therapeutic strategies:

- Direct restoration of tumor suppressor function
- Targeting pathways that become essential when tumor suppressors are lost
- Exploiting synthetic lethal interactions
- Restoring downstream effector functions

Future Directions

The study of tumor suppressors continues to evolve with new technologies and discoveries:

- Single-cell technologies are revealing the heterogeneity of tumor suppressor loss within tumors
- Epigenetic mechanisms of tumor suppressor inactivation are becoming better understood
- Liquid biopsies may allow monitoring of tumor suppressor status during treatment

- Combination therapies that target multiple tumor suppressor pathways are being developed

Conclusion: The Balance of Power

The tumor suppressors we've discussed today—TP53, APC, and RB—represent some of our most important defenses against cancer. They work through different mechanisms and protect against different aspects of malignant transformation, but they share the common function of maintaining cellular integrity and preventing uncontrolled growth.

The fact that these proteins are so frequently inactivated in cancer tells us how crucial they are for cancer prevention. Understanding their normal functions and how they're lost in cancer has been fundamental to our understanding of cancer biology and has opened up important therapeutic opportunities.

As we move forward in this course, we'll see how the loss of tumor suppressor function interacts with oncogene activation, genomic instability, and other cancer-promoting mechanisms to drive the complex process of malignant transformation.

The balance between oncogenes and tumor suppressors ultimately determines whether cells remain normal or become cancerous. Disrupting this balance—through oncogene activation, tumor suppressor loss, or both—is at the heart of cancer development.

Next time, we'll explore the enabling mechanisms that facilitate the acquisition of cancer hallmarks, focusing on genomic instability and tumor-promoting inflammation. We'll see how these enabling characteristics create the conditions that allow oncogene activation and tumor suppressor loss to drive cancer development.

Thank you for your attention, and I look forward to our discussion.